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المملكة المغربية
وزارة التربية الوطنية
والتعليم الأولي والرياضة

Cadres de référence de l'examen national du baccalauréat
Options internationales du baccalauréat marocain - 2025
- option : anglais -
Subject : MATHEMATICS
Section : Experimental Sciences
Filière : Physics and Life -Earth Sciences

Royaume du Maroc * Ministère de l'Education Nationale, du Préscolaire et des Sports
Centre National des Examens Scolaires et de l'Evaluation des Apprentissages

Centre National des examens scolaires et de l'Evaluation des apprentissages
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1. First main domain: Analysis

1.1. First subdomain: Numerical Sequences

1.1.1. Use geometric sequences and arithmetic sequences to study examples of sequences in the form:

$$u_{n+1} = au_n + b \quad \text{or} \quad u_{n+1} = \frac{au_n + b}{cu_n + d}$$

1.1.2. Use the limits of the reference sequences and the convergence criteria to determine the limits of a numerical sequence.

1.1.3. Determine the limit of the composite of a numerical sequence and a continuous function (Sequence in the form $v_n = f(u_n)$)

1.1.4. Study the convergence of a sequence (u_n) in the form $u_{n+1} = f(u_n)$ where f is a continuous function on an interval I verifying $f(I) \subset I$ and determine its limit.

1.1.5. Use numerical sequences to solve various problems from different fields.

1.2. Second subdomain: Continuity, differentiation, study of functions and calculus of integral

1.2.1. Study the continuity of a numerical function at a point using limits calculation.

1.2.2. Determine the image of an interval or a segment by a continuous function or by a continuous and strictly monotonic function.

1.2.3. Apply the Intermediate Value Theorem to study some equations and inequations or to study the sign of some expressions…

1.2.4. Apply the Intermediate Value Theorem, in the case of a continuous and strictly monotonic function on an interval, to prove the uniqueness of the solution of the equation: $f(x) = \lambda$

1.2.5. Study the differentiability of a numerical function at a point and on an interval.

1.2.6. Determine the derivative function of a numerical function.

1.2.7. Determine the monotonicity of a function.

1.2.8. Determine the sign of a function using its variations table.

1.2.9. Determine the sign of a function from its graphical representation.

1.2.10. Solve graphically equations in the form $f(x) = g(x)$ and inequations in the form $f(x) \leq g(x)$

1.2.11. Determine the derivative and the monotonicity of the inverse function of a continuous and strictly monotonic function on an interval and represent it graphically.

1.2.12. Solve application problems about minimum values and maximum values.

1.2.13. Use the first derivative and the second derivative to study a numerical function and to prove some inequalities…

1.2.14. Determine the primitive functions of usual functions.

1.2.15. Use derivation formulas to determine the primitive functions of a function on an interval.

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1.2.16. Master the algebraic calculation on Logarithms.

1.2.17. Master and solve logarithmic equations, inequalities and systems.

1.2.18. Recognize and apply the decimal logarithm (in particular to solve equations in the form $10^x = a$ and inequalities in the form $10^x \leq a$ or $10^x \geq a$)

1.2.19. Master and apply the basic logarithmic limits.

1.2.20. Master and solve equations, inequalities and systems that contain Napierian exponentials.

1.2.21. Master and apply the basic limits of the Napierian exponential function.

1.2.22. Study and represent graphically functions and composite functions among the functions included in the syllabus (domain of definition, elements of symmetry, periodicity, monotonicity, infinite branches, tangent lines, concavity, inflexion points.)

1.2.23. Solve the differential equation of type $y' = ay + b$, where $a \in \square$ and $b \in \square$

1.2.24. Solve the differential equation of type $y'' + ay' + by = 0$, where $a \in \square$ and $b \in \square$

1.3. Third subdomain: Calculating integrals.

1.3.1. Use a primitive function or the technique of integration by parts in order to calculate the integral of a function.

1.3.2. Use the integral properties.

1.3.3. Calculate the domain's area delimited by two curves in the plane.

1.3.4. Calculate the volume of the solid of revolution generated by the rotation of the curve of a function about the x-axis.

2. Second main domain: Algebra and Geometry

2.1. First subdomain: Scalar product in V_3

2.1.1. Express and prove the orthogonality of two vectors using the Scalar product.

2.1.2. Express vectorially the orthogonality and its properties.

2.1.3. Express analytically the orthogonality and its properties.

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2.2. Second subdomain: Applying the scalar product in space.

2.2.1. Determine an equation of a plane defined by a point and a normal vector.

2.2.2. Determine a parametric representation of the straight line passing through a point and orthogonal to a plane.

2.2.3. Study the set of points $M(x, y, z)$ such that: $x^2 + y^2 + z^2 + ax + by + cz + d = 0$

2.2.4. Determine a cartesian equation of a sphere defined by its centre and its radius.

2.2.5. Recognize the set of points M in space verifying the relation $\overrightarrow{MA}.\overrightarrow{MB} = 0$

2.2.6. Use the distance of a point to a plane to solve geometrical problems (relative positions of a plane and a sphere and of a straight line and a sphere …).

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2.3. Third Subdomain: cross product

2.3.1. Calculate the area of a triangle using the cross product.

2.3.2. Determine the equation of a plane defined by three non collinear points.

2.3.3. Use the distance of a point to a straight line to solve geometrical problems.

2.3.4. Apply the cross-product to solve geometrical problems

2.4. Fourth Subdomain: Complex numbers

2.4.1. Master algebraic calculations on the complex numbers (In their writings: algebraic, trigonometric, and exponential)

2.4.2. Go from the algebraic form to the trigonometric form of a complex number and inversely.

2.4.3. Linearize trigonometric monomials using the exponential form of a complex number.

2.4.4. Interpret, using the complex tool, the following geometrical concepts: distance between two points, measurement of angles, collinearity of points, collinearity, and orthogonality of vectors.

2.4.5. Express the translation, the homothety and the rotation using complex tools.

2.4.6. Recognize a translation, homothety or rotation from their complex expressions.

2.4.7. Using complex numbers to solve geometric problems (collinearity, orthogonality…)

2.4.8. Solve the equation $az^2+bz+c=0$ in the complex numbers set where a, b and c are real numbers.

2. 4.9. Solve equations which lead to a second-degree equation with one variable and real coefficients.

2.5. Fifth subdomain: calculating probabilities.

2.5.1. Using the appropriate counting model according to the studied situation.

2.5.2. Calculate the probability of the union of two events, the probability of the opposite event of an event and the probability of the intersection of two events.

2.5.3. Calculate the conditional probability and apply it to determine the probability of the intersection of two events.

2.5.4. Recognise the independence of two events.

2.5.5. Determine the probability law of a random variable and calculate its diverse part

2.5.6. Recognize the binomial law and apply it in various situations.

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3. Specification table

A. According to the main domains

Main Domain	Subdomains	Importance rate
Analysis	Numerical Sequences	55%
	Continuity, differentiability, and study of functions	
	Calculating integrals	
Algebra and Geometry	Scalar product in V_3	15%
	Applications of the scalar product in space	
	cross product	
	Complex Numbers	30%
	Calculating probabilities	
Total		100%

B. According to the skill levels

Skill levels	Importance rate
Direct Application of knowledge (definition, property, algorithm, formula, technique, rule...).	50%
Evoke and apply non-explicit knowledge in a question (definition, property, theorem, algorithm, expression, technic, rule...) in familiar situations.	35%
Deal with unfamiliar situations using knowledge synthesis and results.	15%

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